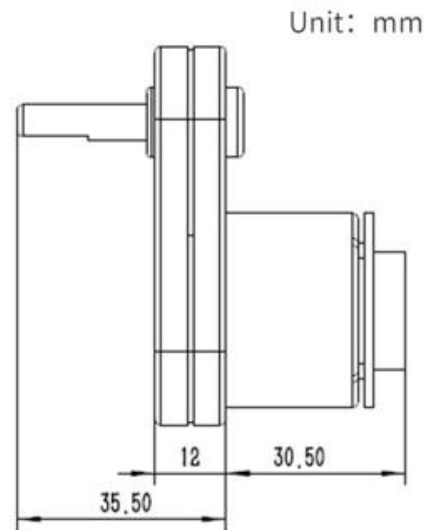
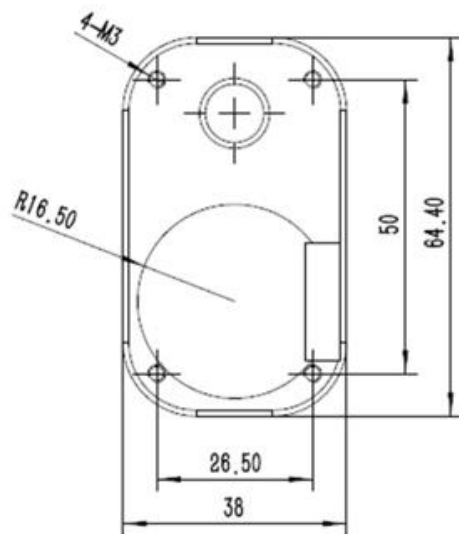


Specifications

Motor output shaft: 6mm diameter D-type shaft

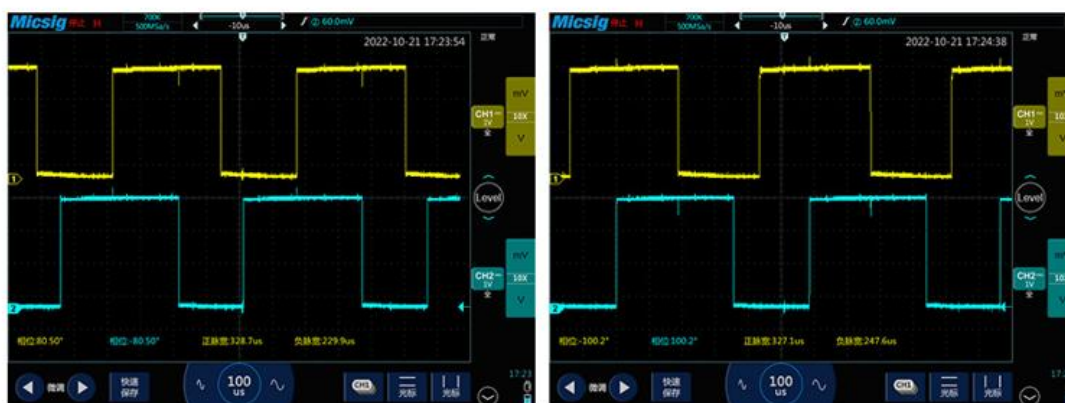


Motor model	L-type 520 encoder motor	Motor rated voltage	12V
Motor type	Permanent magnet brush	Output shaft	D-type eccentric shaft with 6mm diameter
Stalled torque	10kg·cm	Rated torque	4.4kg·cm
Speed before deceleration	12000rpm	Speed after deceleration	300rpm $\pm$ 5%
Rated power	6W	Stalled current	4A
Rated current	0.5A	Gear set reduction ratio	1:40
Encoder type	AB incremental Hall encoder	Encoder supply voltage	3.3V
Number of magnetic ring lines	11 lines	Interface type	PH2.0-6P
Function	MCU can directly read signal pulses	Weight	161g

Encoder Output Description

The phase difference between the two signals is 100°, and the direction of motor rotation can be determined based on the order of the two signals. The current tire travel distance can be calculated based on the number of signal pulses per unit time and the tire circumference. If only the number of AB phase pulses per unit time is detected, the current motor speed can also be measured.

Take a 1:30 reduction ratio motor as an example. The motor outputs 11 pulses per single phase when the motor rotates one circle. With a 1:30 reduction ratio, the maximum output of the motor output shaft is $(30 \times 11 \times 4 =) 1320$ counts per rotation. The phase difference between the AB two-phase output pulse signal is 100°, which can detect the direction of motor rotation.



Forward

Reverse

Encoder Parameters					
Model	Encoder line number	Type	Supply voltage	Encoder protection	Applicable MCU
Hall encoder	11ppr	Magnetic induction	3.3V	Exposed (magnetic encoder is more stable and does not require a back cover)	Almost all MCU

Wiring Instructions

- 1: Motor power line +
- 2: Motor power line -
- 3: Sensor signal negative
- 4: Sensor signal positive 3.3V
- 5: Sensor signal line B phase
- 6: Sensor signal line A phase

